

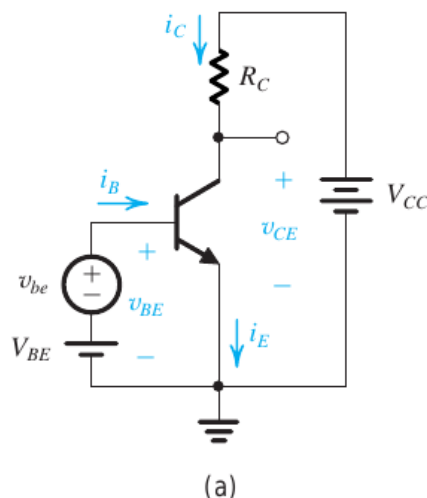
Lecture 12

EE-215 Electronic Devices and Circuits

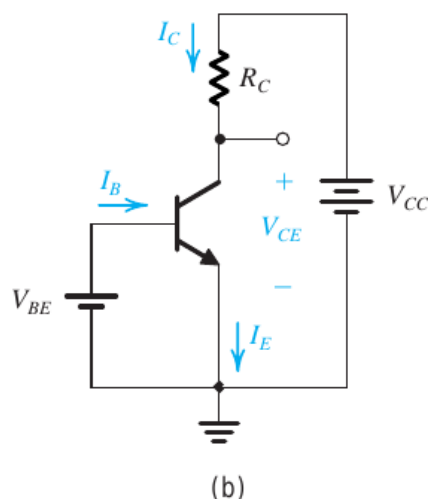
Asst Prof Muhammad Anis Ch

Small-Signal Operation and Models

- Thus for a BJT amplifier, linear amplification can be achieved by
 - biasing the BJT to operate in the active region
 - and by keeping the input signal small
- for the conceptual amplifier circuit shown in fig
 - here the base-emitter junction is forward-biased by a dc voltage V_{BE}
 - the reverse-bias of the collector-base junction is achieved by connecting
 - the collector to another power supply of voltage V_{CC} through a resistor R_C
 - the input signal to be amplified v_{be} is superimposed on the dc bias voltage V_{BE}
 - let's first consider the dc bias conditions by setting the signal $v_{be} = 0$



- thus the dc currents and voltages can be given as
 - $I_C = I_S e^{V_{BE}/V_T}$
 - $I_E = I_C / \alpha$
 - $I_B = I_C / \beta$
 - $V_{CE} = V_{CC} - I_C R_C$
 - Note that for active-mode operation, V_C should be greater than $(V_B - 0.4)$ by an amount
 - that allows for the required signal swing at the collector



The Collector Current and the Transconductance

- Now if a signal v_{be} is applied, the total instantaneous base-emitter voltage is

- $v_{BE} = V_{BE} + v_{be}$

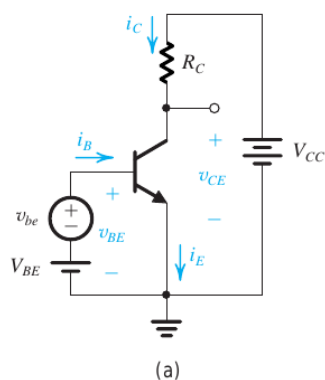
- and the total instantaneous collector current i_C is

- $i_C = I_S e^{v_{BE}/V_T} = I_S e^{(V_{BE} + v_{be})/V_T}$

- $i_C = I_S e^{V_{BE}/V_T} e^{v_{be}/V_T}$

- As $I_C = I_S e^{V_{BE}/V_T}$

- $\Rightarrow i_C = I_C e^{v_{be}/V_T}$



- As in terms of series, exponential can be represented as

- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$

- $\Rightarrow i_C = I_C e^{v_{be}/V_T} = I_C \left(1 + \frac{v_{be}}{V_T} + \frac{1}{2!} \left(\frac{v_{be}}{V_T} \right)^2 + \dots \right)$

- $i_C = I_C e^{v_{be}/V_T} = I_C \left(1 + \frac{v_{be}}{V_T} + \frac{1}{2!} \left(\frac{v_{be}}{V_T} \right)^2 + \dots \right)$

- now if the amplitude of the signal v_{be} is kept sufficiently small i.e. $v_{be} < V_T$ or

- $\frac{v_{be}}{V_T} < 1 \Rightarrow \left(\frac{v_{be}}{V_T} \right)^2 < \frac{v_{be}}{V_T}$

- thus we can retain only the 1st two terms, when $v_{be} < V_T$

- $\Rightarrow i_C = I_C \left(1 + \frac{v_{be}}{V_T} + \frac{1}{2!} \left(\frac{v_{be}}{V_T} \right)^2 + \dots \right) \approx I_C \left(1 + \frac{v_{be}}{V_T} \right)$

- this is the small signal approximation.

- and under this approximation, the total collector current is

- $i_C \approx I_C \left(1 + \frac{v_{be}}{V_T} \right) = I_C + \frac{I_C}{V_T} v_{be}$

- i.e. the collector current is composed of the dc bias value I_C and a signal

component i_c ,

■ where $i_c = \frac{I_C}{V_T} v_{be}$

• $i_c = \frac{I_C}{V_T} v_{be}$

◦ thus the signal current in the collector is proportional to the corresponding base-emitter signal voltage

■ i.e. $i_c = g_m v_{be}$ where $g_m = \frac{I_C}{V_T}$

◦ g_m is called the transconductor and is given as $g_m = \frac{I_C}{V_T}$

◦ thus the transconductance of the BJT is directly proportional to the collector bias current I_C

◦ Note that BJTs have relatively high transconductance as compared to MOSFETs (for a MOSFET, $g_m = \frac{I_D}{V_{OV}/2}$) e.g. at $I_C = 1mA$, $g_m = 40mA/V$

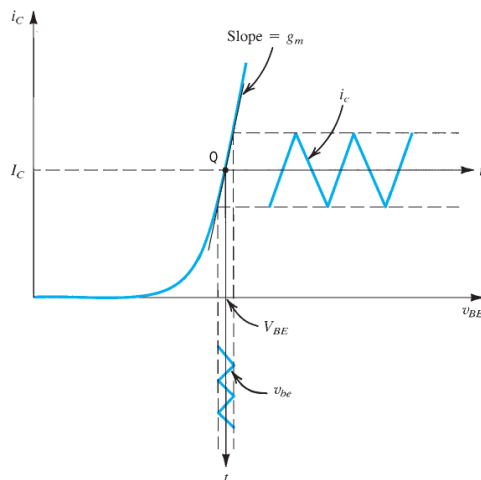
• Graphical interpretation for g_m is as shown in figure

◦ Note that g_m is equal to the slope of the $i_C - v_{BE}$ characteristic at the bias point Q

■ i.e. $g_m = \left. \frac{\partial i_C}{\partial v_{BE}} \right|_{i_C=I_C}$

■ as $i_C = I_S e^{v_{BE}/V_T}$

■ $g_m = \left. \frac{\partial i_C}{\partial v_{BE}} \right|_{i_C=I_C} = \left. \frac{\partial}{\partial v_{BE}} [I_S e^{v_{BE}/V_T}] \right|_{i_C=I_C} = I_S \left. \frac{\partial}{\partial v_{BE}} [e^{v_{BE}/V_T}] \right|_{i_C=I_C}$

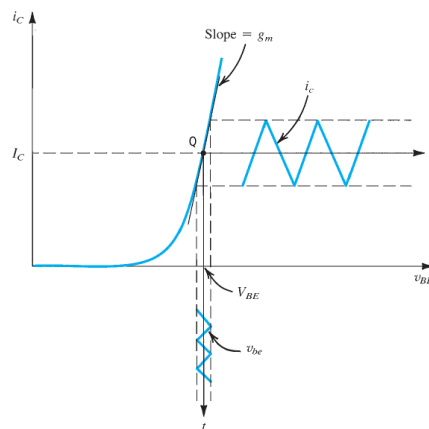


◦ $g_m = I_S e^{v_{BE}/V_T} \left. \frac{\partial}{\partial v_{BE}} [v_{BE}/V_T] \right|_{i_C=I_C} = i_C \left. \frac{\partial}{\partial v_{BE}} [v_{BE}/V_T] \right|_{i_C=I_C}$

• $g_m = \left. \frac{i_C}{V_T} \right|_{i_C=I_C} = \frac{I_C}{V_T}$

◦ thus the small-signal approximation implies

- keeping the signal amplitude sufficiently small that
- operation is restricted to an almost-linear segment of
- the $i_C - v_{BE}$ exponential curve.



The Base Current and the Input Resistance at the Base

- to determine the resistance seen by v_{be} , first we need to determine the total base current i_B
 - as $i_B = \frac{i_C}{\beta}$
 - using $i_C = I_C + \frac{I_C}{V_T} v_{be}$
 - $\Rightarrow i_B = \frac{i_C}{\beta} = \frac{I_C + \frac{I_C}{V_T} v_{be}}{\beta} = \frac{I_C}{\beta} + \frac{1}{\beta} \frac{I_C}{V_T} v_{be}$
 - or $i_B = I_B + i_b$
 - where $I_B = \frac{I_C}{\beta}$ and i_b is the signal component
 - i.e. $i_b = \frac{1}{\beta} \frac{I_C}{V_T} v_{be} = \frac{g_m}{\beta} v_{be} \because g_m = \frac{I_C}{V_T}$
 - thus $i_b = \frac{g_m}{\beta} v_{be}$
 - The small-signal input resistance between base and emitter, looking into the base is denoted by r_π and is defined as
 - $r_\pi = \frac{v_{be}}{i_b}$
- using $i_b = \frac{g_m}{\beta} v_{be}$
 - $r_\pi = \frac{v_{be}}{i_b} = \frac{v_{be}}{\frac{g_m}{\beta} v_{be}} = \frac{\beta}{g_m}$
 - thus r_π is directly proportional to β and is inversely proportional to g_m
 - As $g_m = \frac{I_C}{V_T}$
 - $\Rightarrow r_\pi = \frac{\beta}{g_m} = \frac{\beta V_T}{I_C} = \frac{V_T}{I_B} \because I_B = \frac{I_C}{\beta}$

The Emitter Current and the Input Resistance at the Emitter

- the total emitter current i_E can be given as
 - $i_E = \frac{i_C}{\alpha} = \frac{I_C + i_c}{\alpha} = \frac{I_C}{\alpha} + \frac{i_c}{\alpha}$
 - $\Rightarrow i_E = I_E + i_e$
 - where $I_E = \frac{I_C}{\alpha}$
 - and the signal current $i_e = \frac{i_c}{\alpha}$
 - as $i_c = g_m v_{be} = \frac{I_C}{V_T} v_{be}$
 - $\Rightarrow i_e = \frac{i_c}{\alpha} = \frac{\frac{I_C}{V_T} v_{be}}{\alpha} = \frac{1}{\alpha} \frac{I_C}{V_T} v_{be} = \frac{I_C}{\alpha} \frac{1}{V_T} v_{be} = I_E \frac{1}{V_T} v_{be} \because I_E = I_C / \alpha$
 - thus $i_e = \frac{I_E}{V_T} v_{be}$

- the small-signal resistance between base and emitter looking into the emitter is denoted by r_e and is defined as

$$\blacksquare r_e = \frac{v_{be}}{i_e}$$

$$\bullet r_e = \frac{v_{be}}{i_e}$$

- as $i_e = \frac{I_E}{V_T} v_{be}$

$$\blacksquare \Rightarrow r_e = \frac{v_{be}}{i_e} = \frac{v_{be}}{\frac{I_E}{V_T} v_{be}} = \frac{V_T}{I_E}$$

- as $I_E = \frac{I_C}{\alpha}$ and $g_m = \frac{I_C}{V_T}$

$$\blacksquare \Rightarrow r_e = \frac{V_T}{I_E} = \frac{\alpha V_T}{I_C} = \frac{\alpha}{g_m}$$

- thus $r_e = \frac{V_T}{I_E} = \frac{\alpha}{g_m}$

r_π and r_e

$$\bullet \text{ thus } r_\pi = \frac{v_{be}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B} \text{ and } r_e = \frac{v_{be}}{i_e} = \frac{V_T}{I_E} = \frac{\alpha}{g_m}$$

- from these relations, we can determine the relation between r_π and r_e

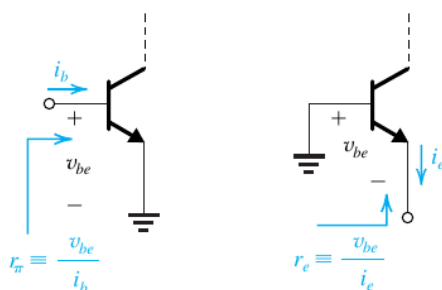


Figure 6.38 Illustrating the definition of r_π and r_e .

- i.e. $v_{be} = i_b r_\pi = i_e r_e$

$$\blacksquare \Rightarrow r_\pi = \frac{i_e}{i_b} r_e$$

- as $\frac{i_e}{i_b} = \frac{i_e}{i_b} \frac{1}{1} = \frac{i_c}{\alpha} \frac{\beta}{i_c} = \frac{i_c(\beta+1)}{\beta} \frac{\beta}{i_c} = \beta + 1$

$$\blacksquare \Rightarrow r_\pi = \frac{i_e}{i_b} r_e = (\beta + 1) r_e$$

Voltage Gain

- by KVL, $V_{CC} = i_C R_C + v_{CE}$

- the total instantaneous collector voltage is

$$\blacksquare v_{CE} = V_{CC} - i_C R_C$$

- under the small signal condition

$$\blacksquare i_C = I_C + i_c$$

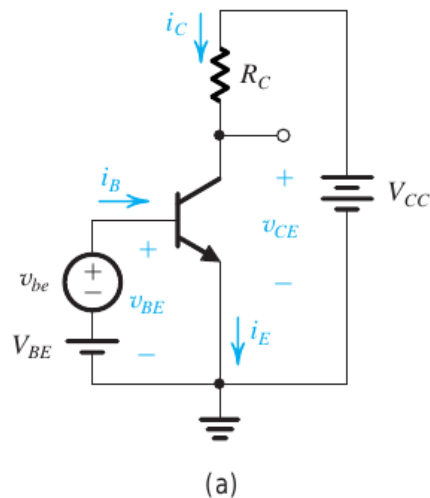
$$\blacksquare \Rightarrow v_{CE} = V_{CC} - i_C R_C = V_{CC} - (I_C + i_c) R_C$$

$$\blacksquare v_{CE} = V_{CC} - I_C R_C - i_c R_C$$

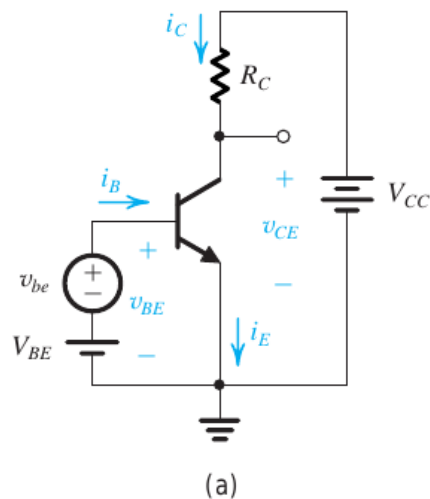
$$\blacksquare v_{CE} = (V_{CC} - I_C R_C) - i_c R_C$$

$$\blacksquare v_{CE} = V_{CE} - i_c R_C \because V_{CE} = V_{CC} - I_C R_C$$

$$\blacksquare \text{ here } V_{CE} \text{ is the dc bias voltage at the collector}$$



- $v_{CE} = V_{CE} - i_C R_C$
 - $\Rightarrow$ the signal component of v_{CE} is
 - $v_{ce} = -i_C R_C$
 - As $i_C = g_m v_{be}$
 - $\Rightarrow v_{ce} = -i_C R_C = -g_m v_{be} R_C$
 - or $v_{ce} = (-g_m R_C) v_{be}$
 - the voltage gain of this amplifier A_v can be given as
 - $A_v = \frac{v_{ce}}{v_{be}} = -g_m R_C$



- Note that the gain is proportional to g_m . As $g_m = \frac{I_C}{V_T}$, thus stable gain can be achieved by making the collector bias current stable
- $A_v = \frac{v_{ce}}{v_{be}} = -g_m R_C$
 - As $g_m = \frac{I_C}{V_T}$
 - $\Rightarrow A_v = \frac{v_{ce}}{v_{be}} = -g_m R_C = -\frac{I_C R_C}{V_T}$
 - here the minus sign indicates that the output signal v_{ce} is 180° out of phase w.r.t. the input signal v_{be}

Separating the Signal and the DC Quantities

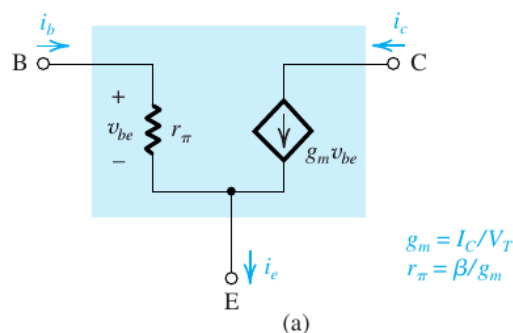
- thus under the small-signal approximation,
 - signal quantities are superimposed on dc quantities

- i.e. every current and voltage in the amplifier circuit is composed of
 - two components: a dc component and a signal component
 - thus $v_{BE} = V_{BE} + v_{be}$, $i_C = I_C + i_c$, $i_E = I_E + i_e$,
 - $i_B = I_B + i_b$ and $v_{CE} = V_{CE} + v_{ce}$
- thus we can simplify analysis by separating “dc calculations” from “small-signal or ac calculations”
- first we perform dc analysis by suppressing all ac sources, and determine the dc operating point
- for ac analysis, we suppress all dc sources,
 - i.e. dc voltage source is replaced by a short circuit
 - a dc current source is replaced by an open circuit

The Hybrid- π Model

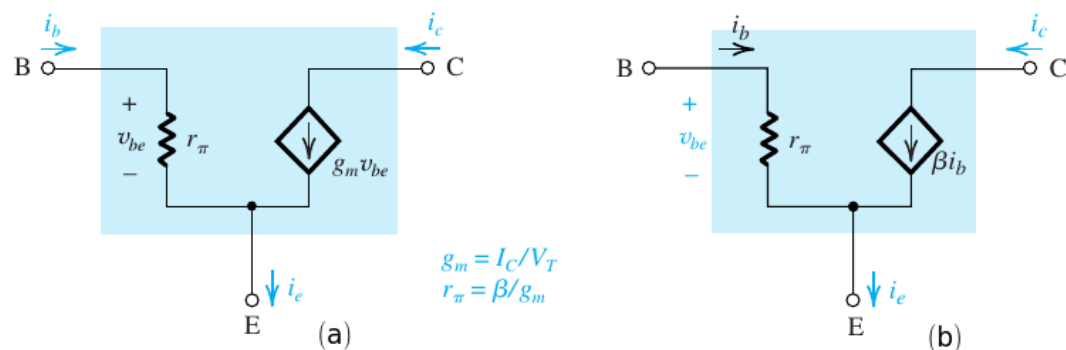
- we have already seen that under the small-signal condition
 - $i_c = g_m v_{be}$ where $g_m = I_C/V_T$
 - $\Rightarrow$ BJT behaves as a voltage controlled current source
 - it accepts a signal v_{be} between base and the emitter
 - and provides a current $g_m v_{be}$ at the collector terminal
 - The small-signal input resistance between the base and emitter, looking into the base is r_π
 - the output resistance is infinite (if base-width modulation effect is neglected)
- thus the small-signal operation of the BJT can be represented by figure

- from this model, $i_c = g_m v_{be}$, $i_b = v_{be}/r_\pi$
- by KCL
 - $i_e = i_b + i_c = \frac{v_{be}}{r_\pi} + g_m v_{be}$
 - $i_e = \left(\frac{1}{r_\pi} + g_m\right) v_{be} = (1 + g_m r_\pi) \frac{v_{be}}{r_\pi}$
 - As $r_\pi = \frac{\beta}{g_m} \Rightarrow g_m r_\pi = \beta$
 - $i_e = (1 + g_m r_\pi) \frac{v_{be}}{r_\pi} = (1 + \beta) \frac{v_{be}}{r_\pi}$
 - but $r_\pi = (\beta + 1)r_e$ or $\frac{r_\pi}{\beta + 1} = r_e$



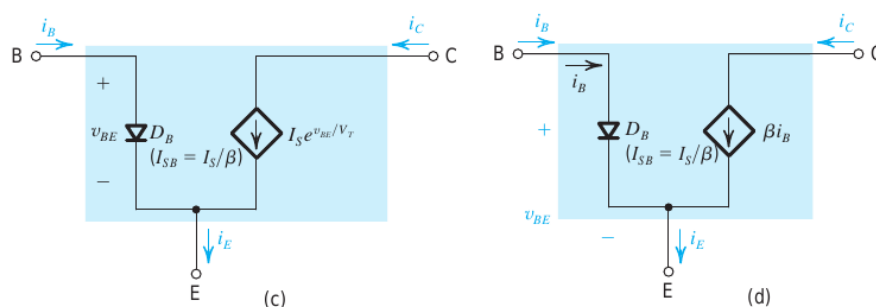
$$\circ \Rightarrow i_e = (1 + \beta) \frac{v_{be}}{r_\pi} = \frac{v_{be}}{r_\pi/(1+\beta)} = \frac{v_{be}}{r_e}$$

- A slightly different equivalent-circuit model is obtained by
 - expressing the current of the controlled source ($g_m v_{be}$) in terms of the base current i_b
 - i.e. $i_c = g_m v_{be} = g_m (i_b r_\pi) = (g_m r_\pi) i_b = \beta i_b \because r_\pi = \beta/g_m$
 - this results in an alternative equivalent-circuit model shown in fig (b)
 - here the transistor is represented by a current-controlled current source, with the control current being i_b



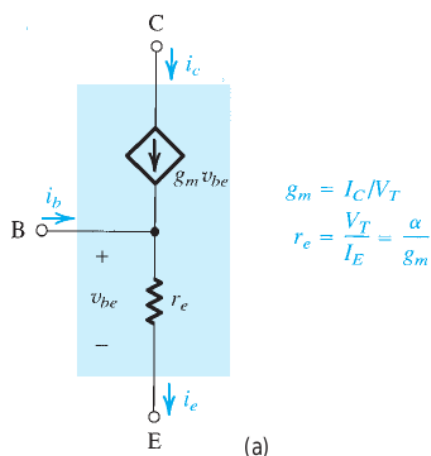
- Note that these models (fig above) are small-signal versions of large signal models (fig below)

◦ Note that r_π is the incremental resistance of D_B



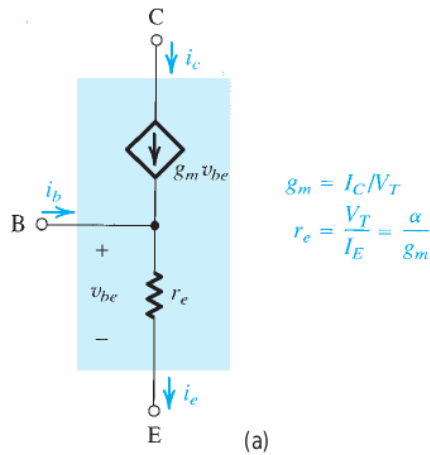
The T Model

- Another possible small-signal model for a BJT called the T-model is shown in figure
 - here the BJT is represented as a voltage-controlled current source
 - with the control voltage being v_{be}
 - here the resistance between base and emitter, looking into the emitter is explicitly shown
 - from the figure we can see that the model yields correct expressions for i_c , i_b and i_e
 - i.e. $i_c = g_m v_{be}$, $i_e = v_{be}/r_e$
 - and $i_b = i_e - i_c = \frac{v_{be}}{r_e} - g_m v_{be} = \left(\frac{1}{r_e} - g_m\right)v_{be}$

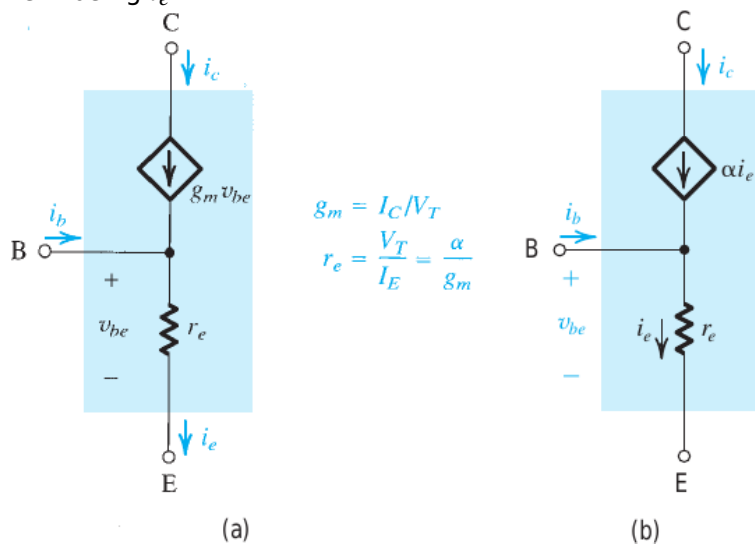


- $i_b = \left(\frac{1}{r_e} - g_m\right)v_{be}$
 - $i_b = \left(\frac{1-g_m r_e}{r_e}\right)v_{be} = \left(\frac{1-\alpha}{r_e}\right)v_{be} \because r_e = \frac{\alpha}{g_m} \text{ or } g_m r_e = \alpha$

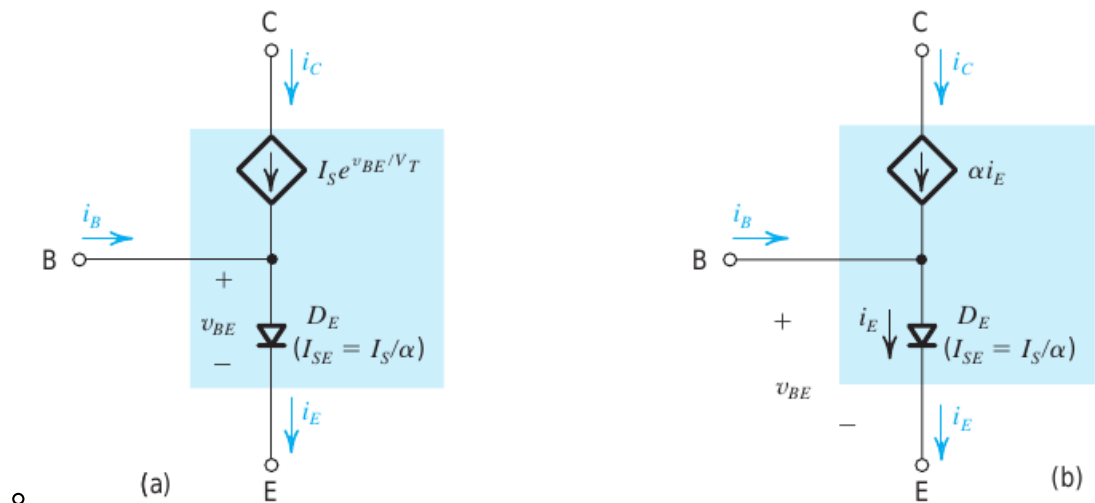
$$\begin{aligned} \blacksquare i_b &= \left(1 - \alpha\right) \frac{1}{r_e} v_{be} = \left(1 - \frac{\beta}{\beta+1}\right) \frac{1}{r_e} v_{be} \\ \blacksquare i_b &= \left(\frac{(\beta+1)-\beta}{\beta+1}\right) \frac{1}{r_e} v_{be} = \left(\frac{1}{\beta+1}\right) \frac{1}{r_e} v_{be} \\ \blacksquare \text{As } r_\pi &= (\beta+1)r_e \\ \blacksquare &\Rightarrow i_b = \frac{1}{r_\pi} v_{be} \end{aligned}$$



- A slightly different equivalent-circuit model is obtained by
 - expressing the current of the controlled source ($g_m v_{be}$) in terms of the emitter current i_e
 - i.e. $i_c = g_m v_{be} = g_m (i_e r_e) = (g_m r_e) i_e = \alpha i_e \because r_e = \alpha / g_m$
 - this results in an alternative equivalent-circuit model (T-model) shown in fig (b)
 - here the BJT is represented by a current-controlled current source, with the control current being i_e

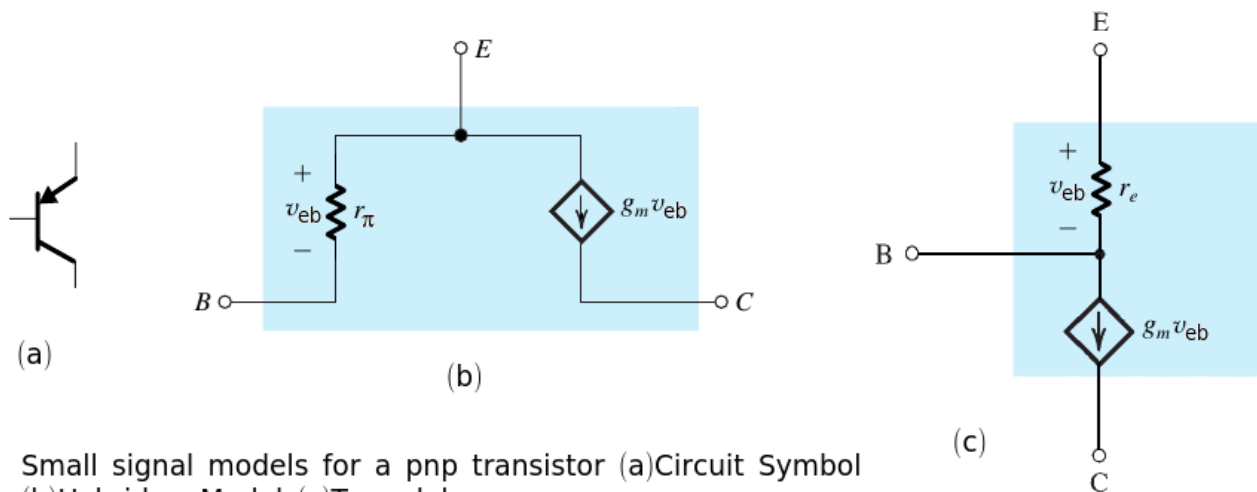


- Note that these small-signal models are small-signal versions of the large-signal models (fig below)
 - and r_e is the incremental resistance of D_E



Small-Signal Models of the pnp Transistor

- for a pnp transistor,



Small signal models for a pnp transistor (a)Circuit Symbol

(b)Hybrid- π Model (c)T model

The base-width modulation effect can be included by placing r_o between C and E

- Recall that for a pnp transistor, larger signal models can be given as

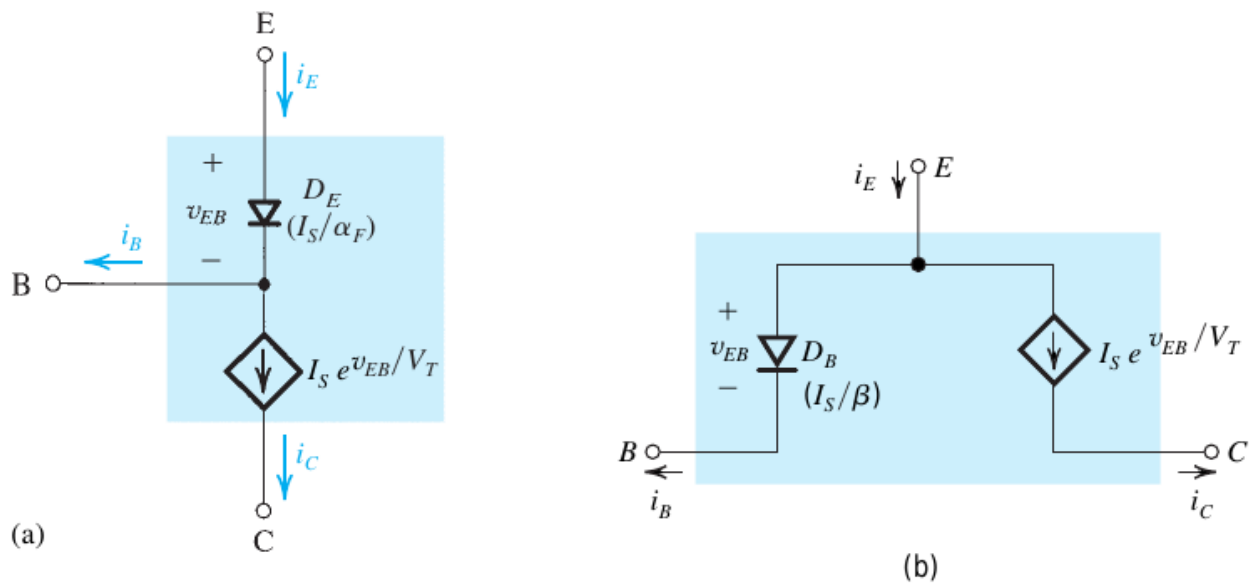
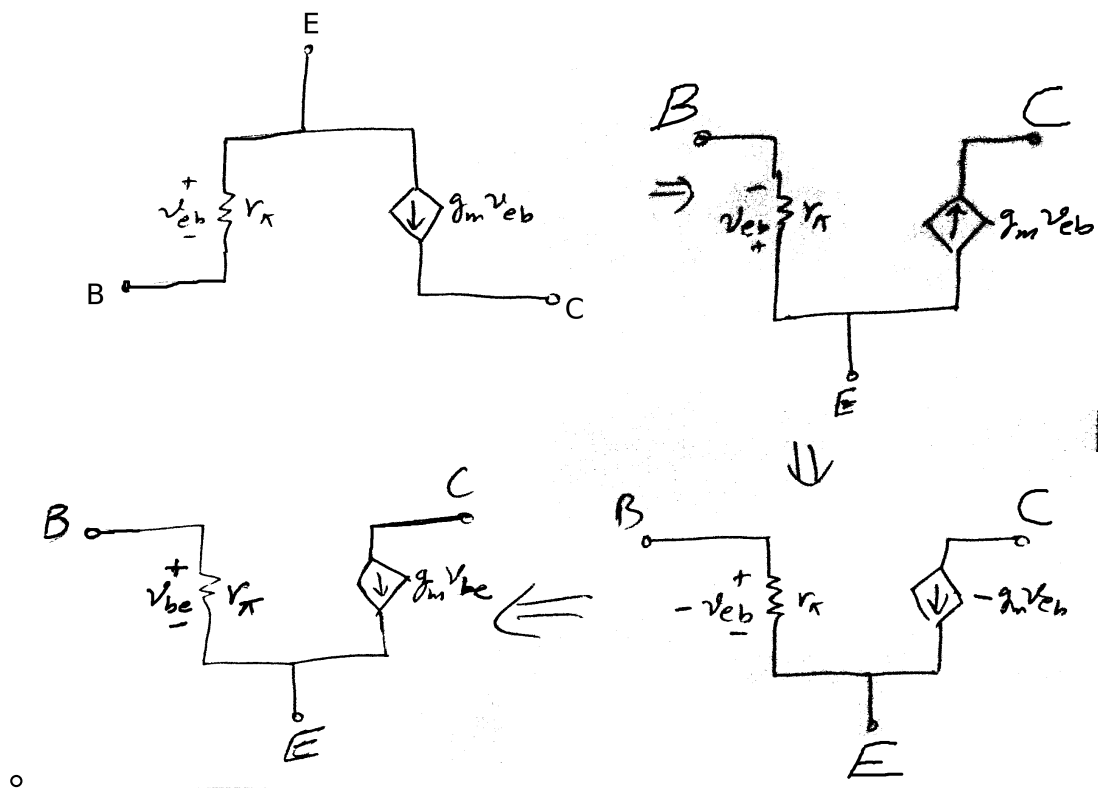
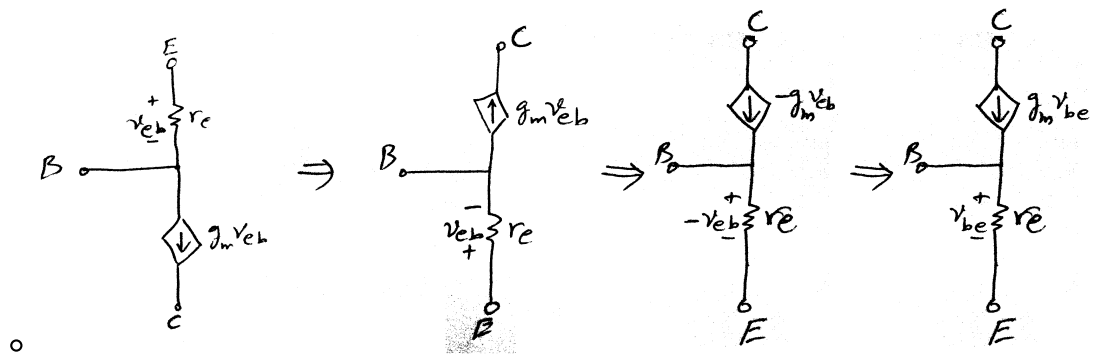


Figure 6.11 Two large-signal models for the *pnp* transistor operating in the active mode.

- Also the above small-signal models developed for npn transistors,
 - apply equally well to the pnp transistor with no change in polarities.
 - to see why?





Augmenting the Small-signal Models to account for the Early Effect

- because of Early effect, i_C depends not only on v_{BE} and also on v_{CE}
 - i.e. $i_C = I_S e^{v_{BE}/V_T} \left(1 + \frac{v_{CE}}{V_A}\right)$
 - As a consequence of Early effect, the output resistance looking into the collector is not infinite and can be given as
 - $r_o = \frac{V_A}{I_C'}$
 - where V_A is the Early voltage and I_C' is the collector current with the Early effect neglected.
 - The Early effect can be included in the small-signal models by placing r_o between the collector and the emitter terminals

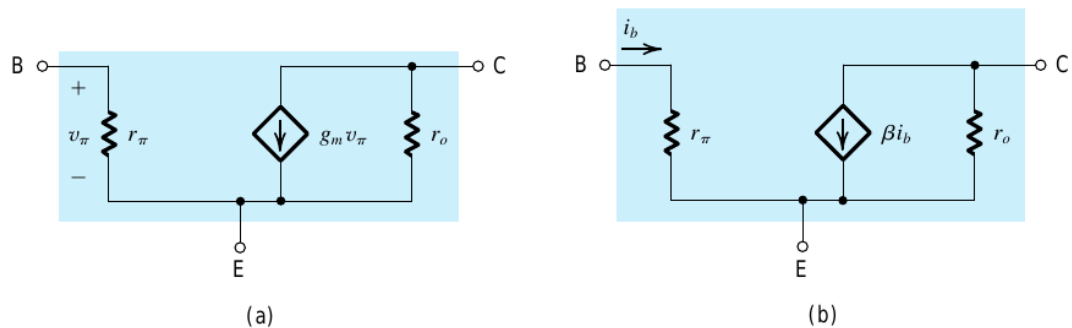
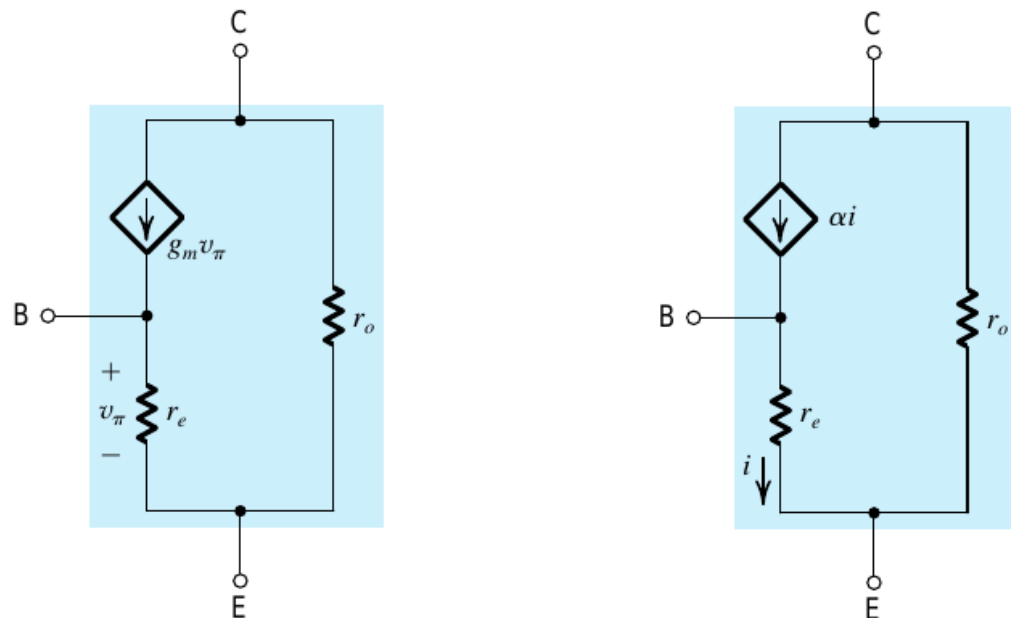


Figure 6.47 The hybrid- π small-signal model, in its two versions, with the resistance r_o included.



Application of the Small-Signal Equivalent Circuits

- Thus in the analysis of a BJT amplifier circuit, we follow the following steps
 1. Suppress the signal sources and determine the dc operating point of the BJT

(particularly the dc collector current, I_C)

2. Calculate the values of the small signal model parameters:

■ i.e. $g_m = \frac{I_C}{V_T}$, $r_\pi = \frac{\beta}{g_m}$, $r_e = \frac{\alpha}{g_m}$

3. Eliminate the dc sources by replacing each dc voltage source with a short-circuit and each current source with an open-circuit

4. Replace the BJT by its small-signal equivalent circuit model

5. Analyse the resulting circuit to determine the required signal quantities e.g. A_v , R_{in} etc.

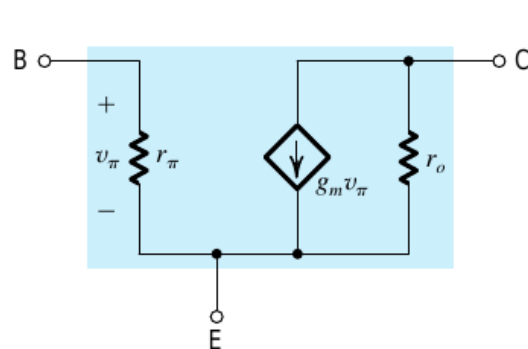
• Table summarizes the small-signal models and relevant relations for a BJT

◦ note that these small-signal models and relations apply equally well to both the npn and the pnp transistors with no change in polarities

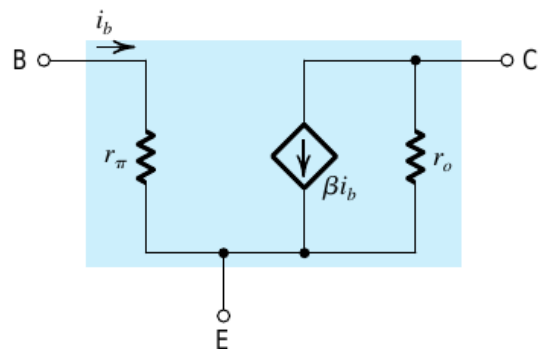
Table 6.4 Small-Signal Models of the BJT

Hybrid- π Model

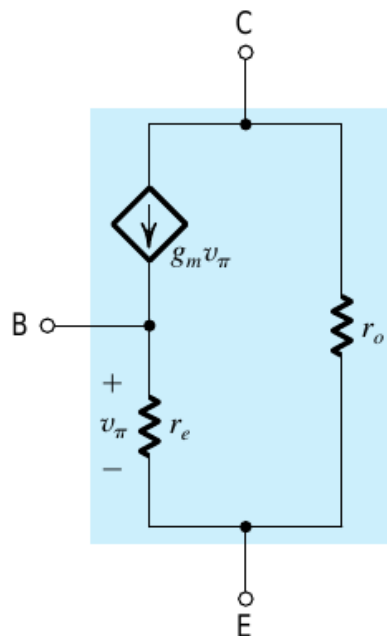
■ $(g_m v_\pi)$ Version



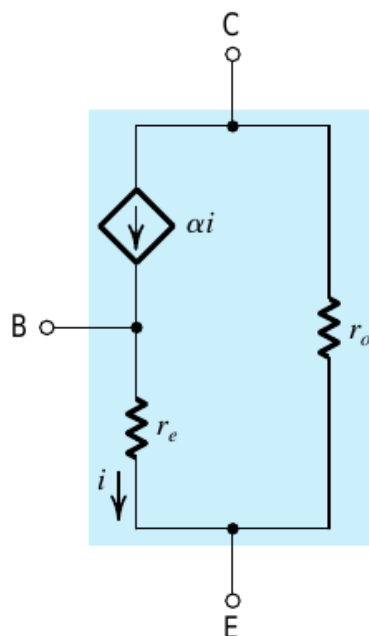
■ (βi_b) Version



■ $(g_m v_\pi)$ Version

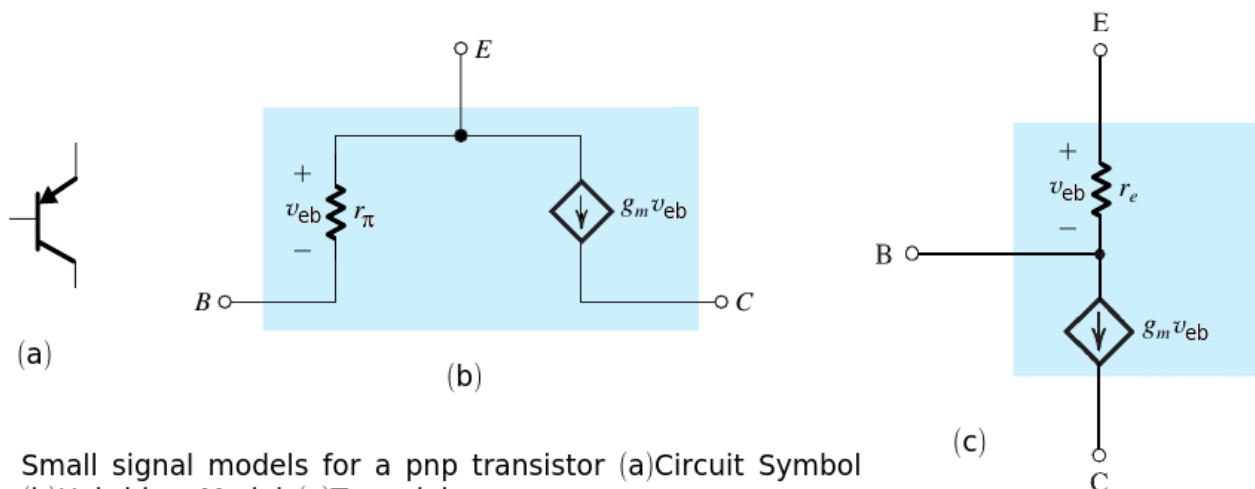


■ (αi) Version



Model Parameters in Terms of DC Bias Currents			
$g_m = \frac{I_C}{V_T}$	$r_e = \frac{V_T}{I_E} = \alpha \frac{V_T}{I_C}$	$r_\pi = \frac{V_T}{I_B} = \beta \frac{V_T}{I_C}$	$r_o = \frac{ V_A }{I_C}$
In Terms of g_m			
$r_e = \frac{\alpha}{g_m}$	$r_\pi = \frac{\beta}{g_m}$		
In Terms of r_e			
$g_m = \frac{\alpha}{r_e}$	$r_\pi = (\beta + 1)r_e$	$g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$	
Relationships between α and β			
$\beta = \frac{\alpha}{1 - \alpha}$	$\alpha = \frac{\beta}{\beta + 1}$	$\beta + 1 = \frac{1}{1 - \alpha}$	

- for a pnp transistor,



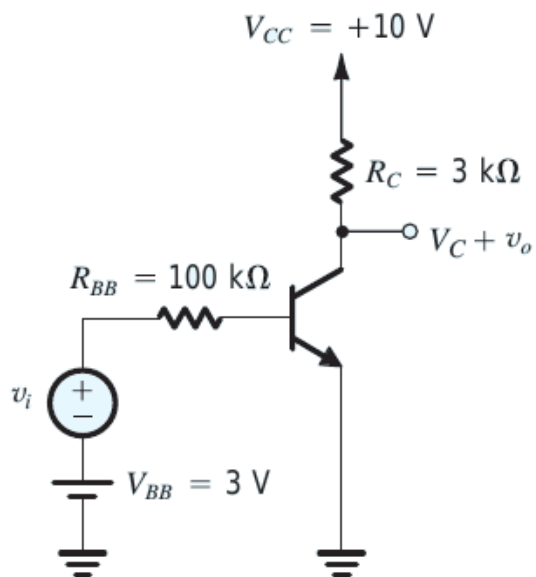
Small signal models for a pnp transistor (a)Circuit Symbol

(b)Hybrid- π Model (c)T model

The base-width modulation effect can be included by placing r_o between C and E

Example 6.14

- Analyze the transistor amplifier shown in Fig 6.42(a) to determine its voltage gain v_o/v_i . Assume $\beta = 100$.



Example 6.16

- Analyze the transistor amplifier shown in Fig 6.44(a) to determine its voltage gain v_o/v_i . Assume $\beta = 100$.

